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MS4226 Risk Management Models

Group Project Report

Diversification in the Hong Kong Stock Market: Mean–Variance vs Equal Weight Portfolios

Group 12

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1. Introduction

The performance of the Hang Seng Index is primarily driven by the financial, technology, real estate and consumer goods. The portfolio of Hang Seng Index faces issues of industry concentration. This report consists of comparing return-risk performance in Equal–weight portfolio and Mean-Variance to analysis the difference of industries concentration by two methods. It further evaluates diversification effects using asset correlation and portfolio internal average correlation. This study provides appropriate allocation recommendations for investors based on empirical findings.

2. Data Collection and Methodology

This section describes the data used in the analysis. The pool focuses on the four core industries in the Hong Kong stock market, including Finance - HSBC Holdings, AIA Group; Technology - Tencent Holdings, Meituan-W; Real Estate - CK Asset, Sun Hung Kai Properties; Consumer Goods - China Resources Beer, Techtronic Industries. Data are collected from Yahoo Finance (Yahoo Finance, 2025).[1] It covers from Jan 2023 to Dec 2025. Daily adjusted closing prices and index returns are adopted for analysis.

Equal–weight portfolio and Mean-Variance(MV) are two of the main components in Portfolio Construction. The EW portfolio maintains fixed equal weights, while the MV portfolio recalculates optimal weights monthly. There are some rules implemented in both methods. One common rule is that calculated weight must be rebalanced monthly. It is realistic to implement and reduces overtrading compared to very frequent re-optimization. For ensuring the portfolio is always comparable across time and avoids unintended leverage or cash positions, all constraints are set to be fully invested ($\sum w = 1$). As short-selling may be infeasible in practice and unconstrained MV solutions can produce unstable offsetting positions, the long-only constraint is also used in the rule ($w \geq 0$).

According to the formula of EW, the weight value for the 8 stocks are equally set to 0.125. Same weighting of each stock is the benchmark for the whole report. The primary objective of the mean-variance portfolio is to maximize the Sharpe ratio. MV optimizations can become overly concentrated due to estimation error. Thus, the weight capacity is set at 0.35 for each asset to improve robustness and enforce diversification. The MV portfolio uses a 252-day rolling estimation window, with an out-of-sample backtest period from January 2024 to December 2025 to avoid look-ahead bias. All rebalancing is conducted at each month-end trading

[1] Yahoo Finance. (2025). *Historical daily adjusted closing prices for Hong Kong-listed equities*. Retrieved April 1, 2026, from https://finance.yahoo.com

day.

The performance analysis covers four dimensions aligned with the research findings. First, for Asset Performance and Correlation, Pearson correlation coefficients of daily returns are presented in a Daily Return Correlation heatmap to analyze asset comovement and diversification potential. Second, for Portfolio Weights and Industry Concentration, sector weight and Top-three industry share graphs, alongside the HHI and effective number of holdings, are adopted to evaluate portfolio concentration. Third, Portfolio Risk and Return Performance is assessed using key metrics including annualized return, volatility, Sharpe ratio, maximum drawdown and turnover for fair risk–return comparison. Finally, InDepth Diversification Analysis utilizes the rolling window portfolio internal average correlation to measure average asset comovement, where lower values indicate better diversification, supported by further turnover evaluation.

3. Empirical Findings

3.1 Asset Performance and Correlation

In this section, the analysis focuses on the performance in weight allocation between Equal-Weight (EW) and Mean-Variance (MV) portfolios. To support the analysis, several analytical pieces of evidence are used.

Figure 1 displays the daily return correlation heatmap. The result shows that correlations are generally moderate (around 0.3-0.7), confirming that the chosen assets are not significantly synchronized. There are relatively high correlations on the Real Estate stocks (CK Assets and Sun Hung Kai Prop.), and the Technology stocks (Tencent and Meituan), indicating the intra-sector industry has stronger correlation.

In contrast, the correlations of cross-sector industries such as HSBC (Finance) with China Resources Beer (Consumer Goods) and Meitun (Technology) are relatively low, representing a weaker relationship between cross-sector industries. Besides, Techtronic is relatively less correlated with the rest, indicating that its daily returns move independently from the three dominant sectors that drive the Hang Seng Index.

To compare the inherent diversification, the Portfolio Internal Average Correlation was calculated. For the Equal-Weight Portfolios of all eight assets, the average value is 0.433. When restricted to the four assets selected by the Mean-Variance optimizer (HSBC, Tencent, CK Assets and Sun Hung Kai Prop), the average value is 0.434. The two portfolios have a low average value (below 0.5), indicating a robust diversification and low volatility. Even though the values are nearly close, the Mean-Variance approach avoids unnecessary concentration within the same industry by assigning zero weight to the more correlated stocks, and identifies a compact subset that maintains essentially the same diversification level as the full universe while eliminating redundant holdings. By favoring low-correlation combinations across sectors, the Mean-Variance Portfolio maintains strong diversification and reduces industry concentration risk.

Figure 2 plots the internal average correlation for both portfolios using a rolling window. The Mean-Variance portfolios frequently exhibit a lower internal correlation than Equal-weight portfolios across much of the sample period, suggesting reduced co-movement in several rebalancing windows. There is a visible gap in mid-2024, where Mean-Variance correlation is highly lower than Equal-weight. However, from mid-2025 onward, the gap narrows significantly, two portfolios become closer and Mean-Variance slightly exceeds Equal-weight at late 2025.

3.2 Portfolio Weights and Industry Concentration

This section is to examine the two portfolio strategies (i.e mean-variance portfolio and equal-variance portfolio) affect the industry concentration and sector diversification in the Hong Kong stock market, and whether the optimization process of the mean-variance portfolio leads to stronger concentration than the equal-weight benchmark.

The equal-weight portfolio serves as the baseline because every stock receives the same allocation by construction, so the strategy automatically maintains broad diversification across the entire asset pool (i.e 12.5% per stock and 25% per industry). However, the mean-variance portfolio produced an uneven distribution of weight. From figure 3, the several holdings reach the maximum weight cap (i.e 35%) which are mainly focused on HSBC, Tencent, CK assets while the other stocks are assigned with very small weights or even close to zero. The “Top drivers vs Other” area chart shows that MV weights are time-varying but consistently dominated by a few key holdings. This indicates that MV performance is primarily driven by a small number of stocks rather than broad exposure across the full universe.

At the industry level, the difference between the two strategies becomes clearer. Figure 4 shows that the equal-weight portfolio keeps capital relatively balanced across the four core industries in the Hong Kong stock market, including Finance, Technology, Real Estate, and Consumer Goods. In contrast, the MV shows a stronger allocation toward Finance, Technology, and Real Estate, while Consumer Goods may receive much less exposure or sometimes no exposure at all in several rebalancing periods based on figure 5. This indicates that MV does not guarantee sector diversification. To quantify diversification at the asset level, we track top-1/top-2 weights and the Herfindahl–Hirschman Index (HHI) over time. EW has constant concentration by construction (HHI = 1/N), whereas MV exhibits substantially higher HHI values, confirming stronger single-name concentration.

To measure this concentration more precisely, we use Top-3 sector share as a direct measure of industry concentration. This allows us to identify whether MV improves performance by concentrating into a small

set of sectors or by maintaining balanced exposure. Based on figure 6, the equal-weight portfolio has a lower concentration level (0.75). By contrast, the mean-variance portfolio shows higher concentration (nearly 1.0) because the optimization process often clusters capital into a smaller number of industries and stocks. This confirms that MV tends to be more industry-concentrated than EW in our setting, reinforcing the trade-off between optimization-driven performance and sector-level diversification. Based on figure 7, we can see that HSBC stock shows the strongest cumulative growth and Tencent shows consistent relative returns. MV optimization naturally tilts toward top-performing Finance/Technology/Real Estate sectors while de-emphasizing Consumer Goods, whereas EW ignores relative performance and maintains balanced exposure.

3.3 Portfolio Risk and Return Performance

This section evaluates and compares the risk-return profiles of the Equal-Weight (EW) and Mean-Variance (MV) portfolios, using the Hang Seng Index (HSI) as a benchmark. In the analysis, we incorporate both standard performance metrics and downside risk measures to provide a broader assessment of the portfolios.

Table 1 shows that the MV portfolio delivers the strongest overall risk-adjusted performance among the three portfolios. It achieves an annualized return of 31.48%, higher than both EW (26.94%) and HSI (30.65%). At the same time, MV records the lowest annualized volatility at 22.29%, compared with 23.90% for EW and 24.50% for HSI. As a result, the MV portfolio attains the highest Sharpe ratio (1.412), exceeding both EW (1.127) and the benchmark (1.251). These results suggest that the optimization procedure is effective in improving the return earned per unit of total risk.

To further assess downside risk, we examine historical VaR and ES at the 95% and 99% confidence levels. At the 95% level, MV performs slightly better than EW, with a lower historical VaR of 1.80% compared with 1.90%, indicating marginally lower exposure to typical daily losses. However, at the 99% level, the historical VaR of MV rises to 3.31%, above EW's 2.93% and HSI's 2.68%. This suggests that although the MV portfolio performs better under normal market conditions, it may remain more sensitive to rare and more extreme loss thresholds.

This apparent weakness is partly moderated by the Expected Shortfall results. At the 99% confidence level, MV reports an ES of 5.88%, slightly below EW (6.04%) and clearly below HSI (6.48%). Therefore, while the MV portfolio crosses a more adverse loss threshold at the 99% VaR level, the average severity of losses beyond that threshold is not worse and is in fact somewhat lower than the alternatives. In other words, the tail-risk evidence is mixed rather than uniformly favorable: MV improves several dimensions of downside risk but does not dominate EW and HSI on every extreme-loss metric.

The VaR exception counts reported in Table 2 are broadly consistent across portfolios and remain close to what would be expected at the selected confidence levels. This provides some support for the calibration of historical risk estimates over the sample.

Overall, the evidence indicates that the MV portfolio offers the most efficient overall balance between return and risk in this study. It outperforms EW and HSI in terms of annualized return, volatility, Sharpe ratio, and maximum drawdown. However, its advantage in downside risk should be interpreted with some caution, since the 99% VaR result points to greater exposure to very extreme loss thresholds. Hence, the main conclusion is not that MV is unambiguously safer in every respect, but rather that it delivers superior overall risk-adjusted performance while keeping a broadly competitive downside risk profile.

3.4 In-Depth Diversification Analysis

This section evaluates the performance of the two different diversification approaches used by each portfolio. Mainly, diversification by increasing the number of stocks (as seen in the EW portfolio vs increasing the number of stocks with low covariance as seen in the MV portfolio).

From section 3.2, we see that MV favours a smaller set of top performing assets. Taking a closer look at these top performers, we see that its diversification lies in the covariance between the assets within the portfolio. As seen from Figure 2, the rolling internal average correlation for the MV portfolio remained lower most of the time as compared to the EW portfolio. As mentioned in Section 3.1, the Portfolio Internal Average Correlation for the Equal-Weight Portfolio is 0.433, and when the four assets selected by the Mean-Variance optimizer (HSBC, Tencent, CK Assets and Sun Hung Kai Prop), the average value is 0.434. Even though both values are almost equal, the MV portfolio hedges across sectors more effectively than the EW portfolio by assigning zero or minimal weights to redundant assets within the same sector. This strategy results in an effective holding size of approximately 3.14 stocks in the MV portfolio as seen in Figure 11. However, this is only the case if the 35% weight cap is implemented. From the same figure, we see that removing the weight cap would essentially reduce the effective holding size of the MV portfolio to half the size, approximately 1.57 stocks instead. While this reduced holding size have resulted in higher performance (annualized return **40.7%**, Sharpe **1.89** as compared to annualized return **31.5%**, Sharpe **1.41**), removing the weight cap essentially causes the portfolio to act as a “1-2 stock portfolio”, which suggests that the cap free MV portfolio path outperforms in sample, but also results in higher fragility and reduced diversification, which is not ideal.

However, the MV portfolio strategy also has to be practical to implement. We see that the capped MV portfolio requires a mean monthly turnover of 13.98% as seen from Table 3. This level of turnover suggests

that it is a realistic strategy to implement as it does not lead to excessive number of transactions, which may increase costs, while still being able to remain responsive to time-varying correlations in the market.

Overall, we see the results of the trade-offs between asset quantity diversification and the risk quality diversification. Despite the MV portfolio (weight capped) being more concentrated than the EW benchmark, it still resulted in a higher performance. Its higher sharpe ratio of 1.41 vs 1.13 suggests that replacing a broad weight diversification of the EW portfolio with a more targeted covariance diversification is safer and more efficient even though it has only ~3 effective holdings compared to 8.

4. Recommendation

Two recommendations are provided to two types of investors. Risk is the main criterion for choosing a different MV strategy.

4.1 For risk-seeking investors

At first glance, Mean-Variance (MV) portfolio is the better option while comparing with Equal-Weight (EW) in Hong Kong market. In table I, MV has a higher return (31.48%), lower volatility (22.29%), and higher Sharpe (1.412). The EW portfolio's return is 5% lower (26.94%), volatility is 1.5% higher (23.90%), and Sharpies 0.3 lower higher (1.127) approximately. However, its result is similar to the Hang Seng index (HSI). With a mean monthly turnover of 13.98% in Table 3, the cost of the transaction fees may reduce the difference between the annual return of MV and HSI. Moreover, MV's 99% VaR is the highest (3.31%). It has the highest possibility of having daily losses frequently. Given these uncertain estimations, investors would rather choose HSI ETF. Therefore, the MV portfolio still needs more improvements.

For the recommendation, it is suggested to have a higher weight capped with 40% or 45%. In figure 3, HSBC always maintains maximum weight (35%) as it has potential to gain great profit to investors over 3 years. Although it results in higher fragility and reduced diversification mentioned in section 3.4, it is worth putting high weight in HSBC. In figure 7, HSBC always leads to the highest index that performs better than other companies, even HSI. With high weight, the proportion of transactions may be reduced. Lower transaction fees may increase the total return of MV. As to the left of the weight, part of it could be the CR Beer. As shown in Figure 1, CR Beer is the lowest correlation to HSBC, against low diversification. As a result, it may attain higher risk and higher 99% VaR but higher potential return. It is suitable for the Risk-Seeking investors who could accept higher tail risk.

4.2 For Risk-Averse Investors

Another problem is the high concentration to Hong Kong market. As Hong Kong has Linked Exchange Rate System, the Hong Kong dollars have fixed relationship to United States Dollar (HKD 7.8≈ USD 1). Hong Kong market would always affect by the trade war of China and United States. In figure 7, all Hong Kong stocks had a significant drop in April 2025 due to the trade war. Therefore, putting all weight in Hong Kong market may not be high diversification and has Systemic risk that market system being unstable.

For the second recommendation, 10% of the portfolio may be replaced as gold or gold ETF (2840.HK), and the cap may be reduced to 32% for the gold. It is another currency that is against HKD and USD that has a low, even negative correlative relationship. In April 2025, the gold price rose significantly and HSI dropped. Therefore, Gold may protect investors when there is a serious financial crash to the market. With 10% gold allocation, it not only helps reduce the 99%VaR in MV, but it also increases the stability of portfolio and diversification.

5. Conclusion

After comparing equal-weight portfolio and mean-variance portfolio in the Hong Kong stock market, this project finds that Mean-variance portfolio has better asset correlation allocation to achieve higher risk-adjusted return and stronger diversification efficiency. Although it exhibits higher industry concentration, it demonstrates greater robustness. A reasonable weight cap could balance return and risk, while adding low-correlation assets can enhance portfolio defensive capability effectively for providing investment guide to risk-tolerant investors.

References

1. *Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77–91.*
2. *Yahoo Finance. (2025). Historical daily adjusted closing prices for Hong Kong-listed equities. Retrieved April 1, 2026, from https://finance.yahoo.com*

Appendix

Figure 1 – Daily Return Correlation

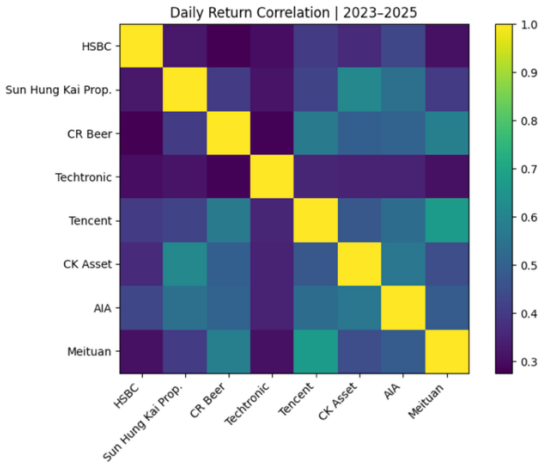


Figure 2 – Portfolio Internal Average Correlation

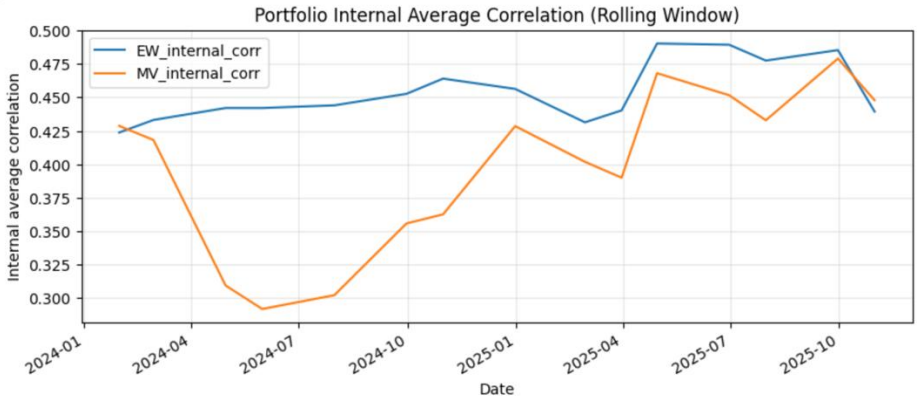


Figure 3 – MV weights over time

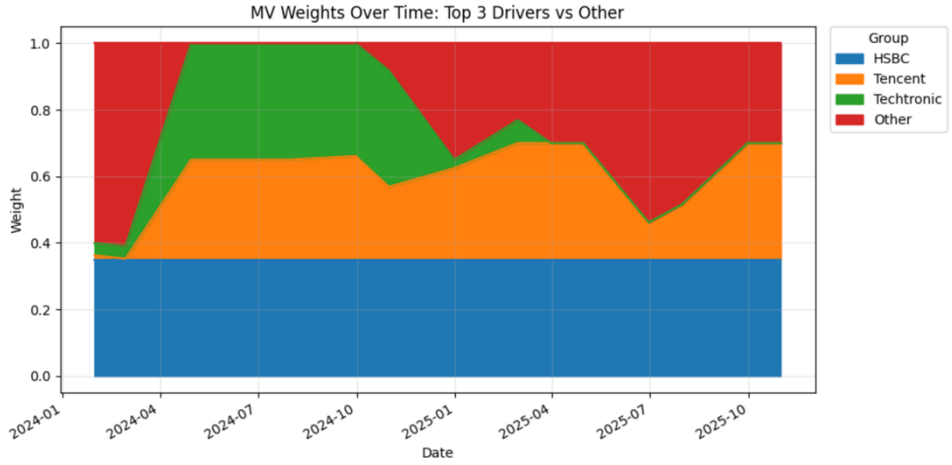


Figure 4 – EW sector weight

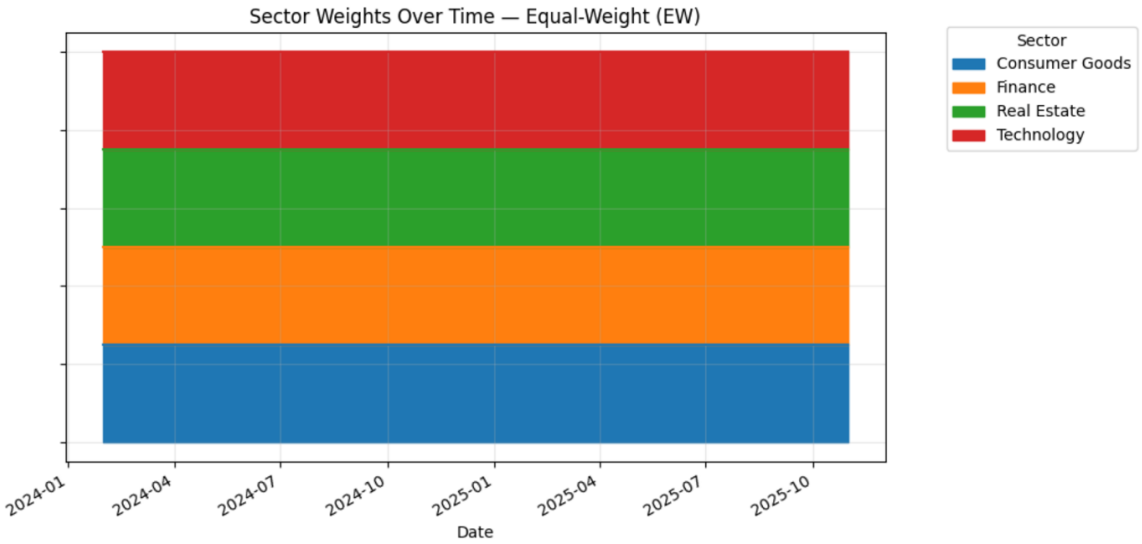


Figure 5 – MV sector weight

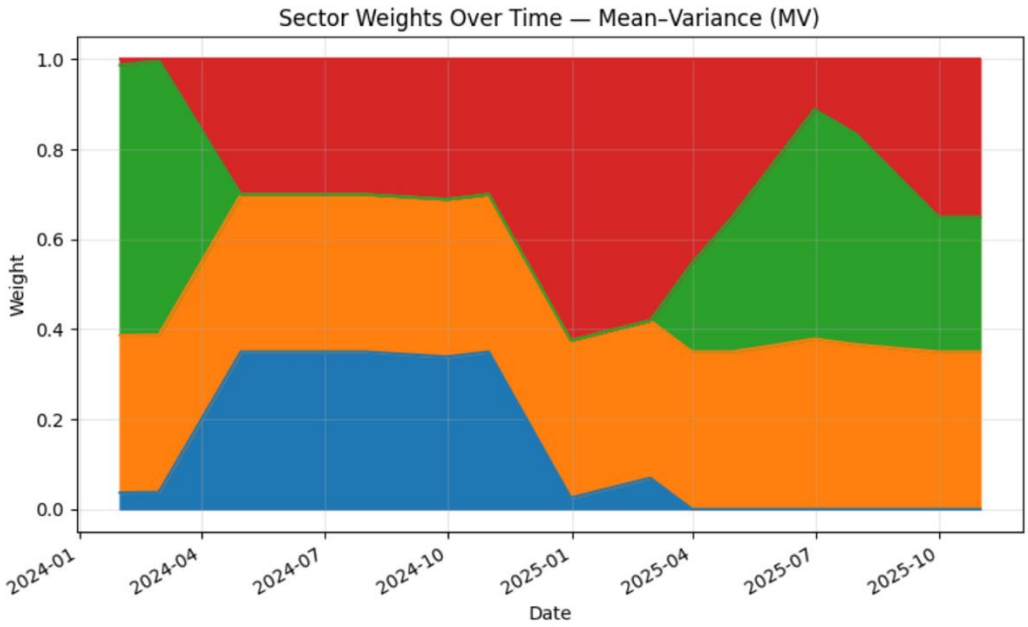


Figure 6 – Industry concentration

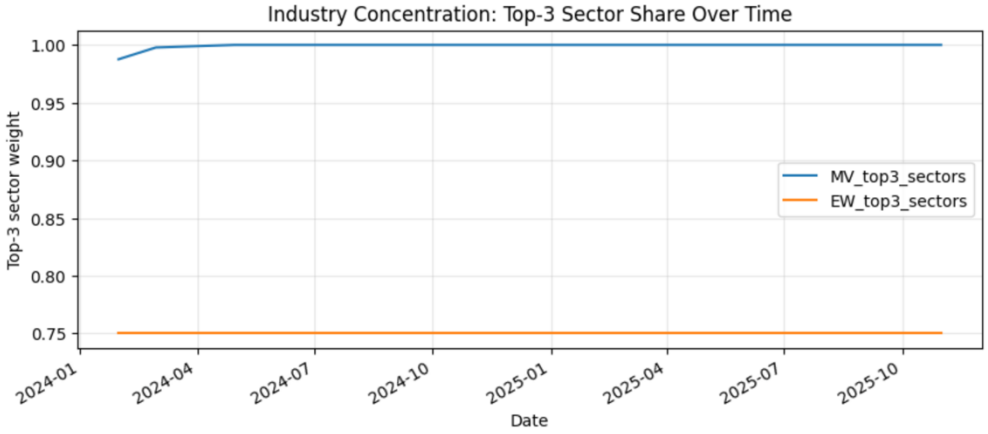


Figure 7 – Normalized adjusted close (log scale)

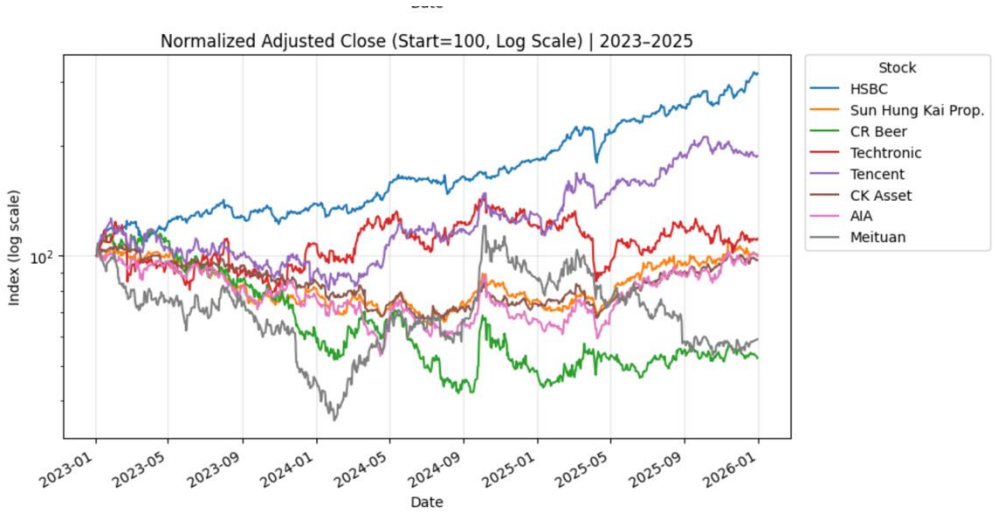


Figure 8 – Distribution of Daily Loss EW

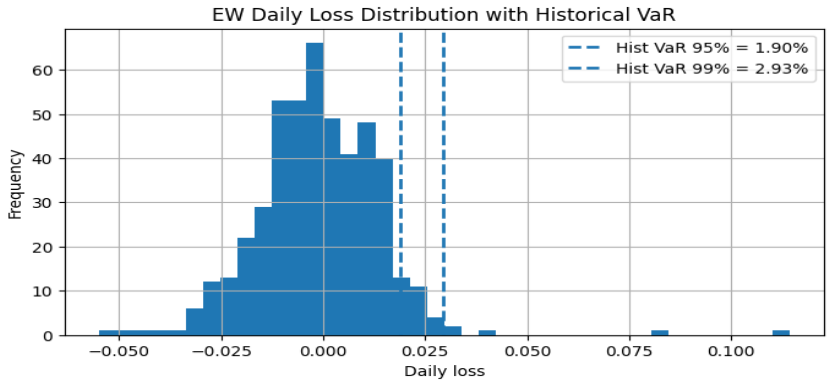


Figure 9 – Distribution of Daily Loss MV

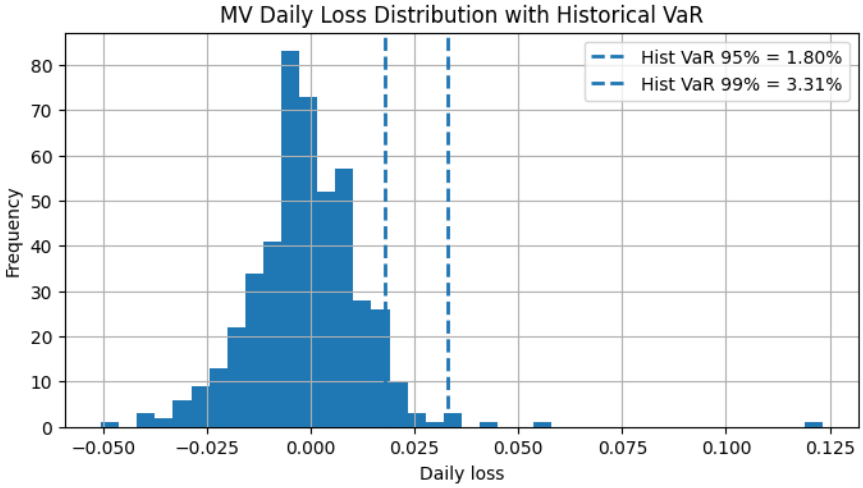


Figure 10 – Distribution of Daily Loss HSI

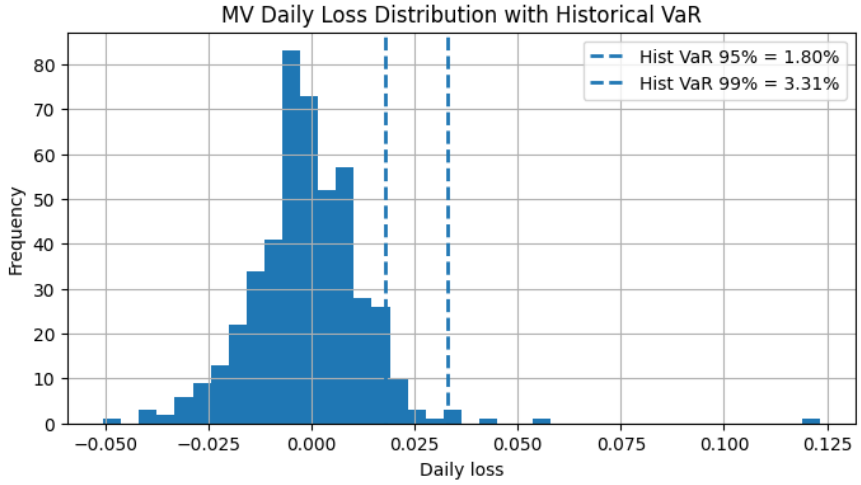


Figure 11 – Effective Holdings Sizes

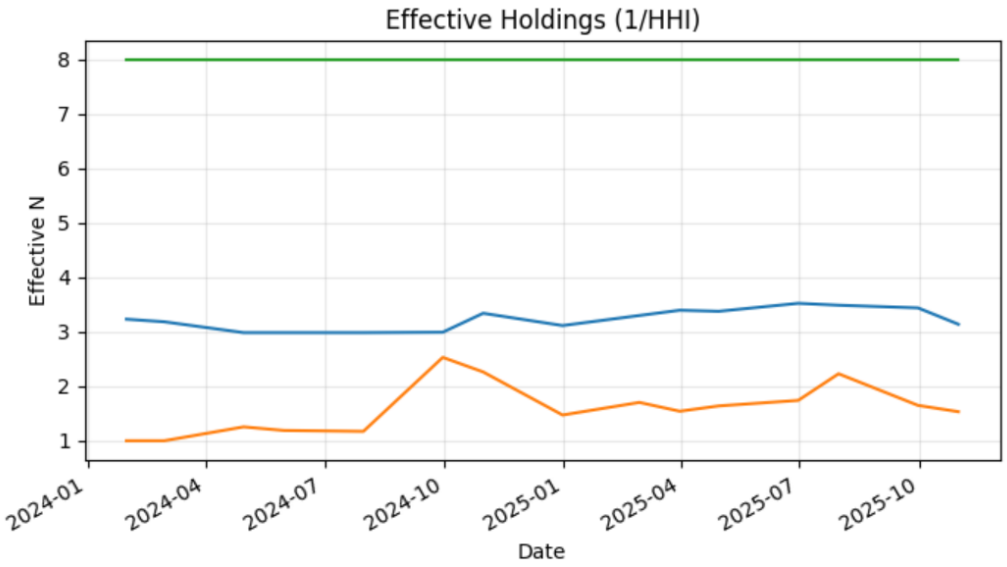


Table 1 – Performance and Risk Summary

Series	Ann Return	Ann Vol	Sharpe	Max Drawdown	Hist VaR 95% (1d)	Hist VaR 99% (1d)	Hist ES 95% (1d)	Hist ES 99% (1d)
EW	26.94%	23.90%	1.127	-21.41%	1.90%	2.93%	3.10%	6.04%
MV (cap 35%)	31.48%	22.29%	1.412	-19.60%	1.80%	3.31%	2.99%	5.88%
Hang Seng (HSI)	30.65%	24.50%	1.251	-19.95%	1.90%	2.68%	3.07%	6.48%

Table 2 – Detailed VaR and Risk diagnostics

Series	Hist VaR 95% (1d)	Hist VaR 99% (1d)	Hist ES 95% (1d)	Hist ES 99% (1d)	Delta-Normal VaR 95% (1d)	Delta-Normal VaR 99% (1d)	Delta-Normal VaR 95% (10d)	Delta-Normal VaR 99% (10d)	Exceptions > Hist VaR 95%	Exceptions > Hist VaR 99%	Obs
EW	1.90%	2.93%	3.10%	6.04%	2.48%	3.50%	7.83%	11.07%	24	5	470
MV (cap 35%)	1.80%	3.31%	2.99%	5.88%	2.31%	3.27%	7.30%	10.33%	24	5	470
Hang Seng (HSI)	1.90%	2.68%	3.07%	6.48%	2.54%	3.59%	8.03%	11.35%	24	5	470

Table 3 – MV Portfolio Monthly Turnover Diagnostics

	MV cap 35%	MV no cap
count	15.000000	15.000000
mean	0.139800	0.144961
std	0.165285	0.136981
min	0.000000	0.000000
25%	0.010605	0.032795
50%	0.099970	0.124860
75%	0.216510	0.233110
max	0.610020	0.386480

Gold Price(2021-2026)



HSI index(2023-2026)


L:25,653.63 C:25,893.54 V:7.79b
25,893.54
25,000.00
22,500.00
20,000.00
17,500.00
7.79B
2023
2024
2026